ITN 7.02 – Case Study

Part 4 (Modules 11–13): Network Design Documentation

Router: NET_1_R1

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1. IPv4 Subnet Scheme (VLSM)

The customer's future-growth network was allocated the address block 192.168.1.0/24. Variable Length Subnet Masking (VLSM) was used to divide this block efficiently, sizing each subnet to match its host requirement while minimising wasted address space.

Segment	Hosts Needed	Subnet Mask	CIDR	Network Address	First Usable	Last Usable	Broadcast
LAN-A	60	255.255.255.192	/26	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
LAN-B	25	255.255.255.224	/27	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95
LAN-C	20	255.255.255.224	/27	192.168.1.96	192.168.1.97	192.168.1.126	192.168.1.127
LAN-D	12	255.255.255.240	/28	192.168.1.128	192.168.1.129	192.168.1.142	192.168.1.143
LAN-E	12	255.255.255.240	/28	192.168.1.144	192.168.1.145	192.168.1.158	192.168.1.159
Link 1	2	255.255.255.252	/30	192.168.1.160	192.168.1.161	192.168.1.162	192.168.1.163
Link 2	2	255.255.255.252	/30	192.168.1.164	192.168.1.165	192.168.1.166	192.168.1.167
Link 3	2	255.255.255.252	/30	192.168.1.168	192.168.1.169	192.168.1.170	192.168.1.171
Link 4	2	255.255.255.252	/30	192.168.1.172	192.168.1.173	192.168.1.174	192.168.1.175

Address range 192.168.1.176 – 192.168.1.255 (80 addresses) remains reserved for future growth.

Design Methodology

- Requirements were sorted from largest to smallest host count (60 → 25 → 20 → 12 → 12 → point-to-point) before assigning blocks, so large subnets are not blocked by leftover fragments.
- Each LAN was assigned the smallest prefix that satisfies its host requirement (usable hosts = 2^n − 2), minimising wasted addresses.
- All four point-to-point links use /30 masks, since a serial/router link only ever needs 2 usable addresses.

2. Physical Network Topology

The network consists of a single core router, NET_1_R1, functioning as the gateway between two internal networks connected via its GigabitEthernet interfaces:

- LAN network (GigabitEthernet0/0/0) — connects to the end-user switch, serving 5 end devices: PC1, PC2, PC3, PC4, and PC5.
- Admin network (GigabitEthernet0/0/1) — connects to the administrative switch, serving 2 end devices: Admin1 and Admin2.

Each end device connects to its respective switch, which uplinks to NET_1_R1. The router performs inter-VLAN/inter-subnet routing between the two IPv6 networks (2001:ABCD:1234:1::/64 and 2001:ABCD:1234:2::/64).

3. Router Configuration Log (CLI Commands)

The following commands were entered on NET_1_R1 via the NET PC1 terminal application to enable IPv6 routing and configure both interfaces:

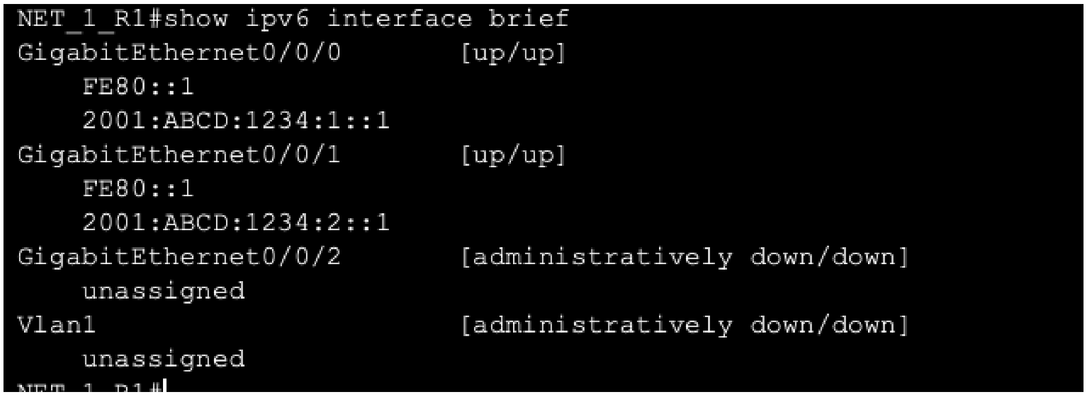
```
Router> enable
Router# configure terminal
Router(config)# hostname NET_1_R1

! Step 2: Enable IPv6 Routing
NET_1_R1(config)# ipv6 unicast-routing

! Configure Interface GigabitEthernet0/0/0
NET_1_R1(config)# interface GigabitEthernet0/0/0
NET_1_R1(config-if)# ipv6 address 2001:ABCD:1234:1::1/64
NET_1_R1(config-if)# ipv6 address FE80::1 link-local
NET_1_R1(config-if)# no shutdown
NET_1_R1(config-if)# exit

! Configure Interface GigabitEthernet0/0/1
NET_1_R1(config)# interface GigabitEthernet0/0/1
NET_1_R1(config-if)# ipv6 address 2001:ABCD:1234:2::1/64
NET_1_R1(config-if)# ipv6 address FE80::1 link-local
NET_1_R1(config-if)# no shutdown
NET_1_R1(config-if)# exit

NET_1_R1(config)# end
NET_1_R1# write memory
Building configuration...
[OK]
```



```
NET_1_R1#show ipv6 interface brief
GigabitEthernet0/0/0       [up/up]
    FE80::1
    2001:ABCD:1234:1::1
GigabitEthernet0/0/1       [up/up]
    FE80::1
    2001:ABCD:1234:2::1
GigabitEthernet0/0/2       [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
NET_1_R1#
```

Screenshot: CLI ตอน config router (NET_1_R1) — show ipv6 interface brief ยืนยันสถานะ up/up

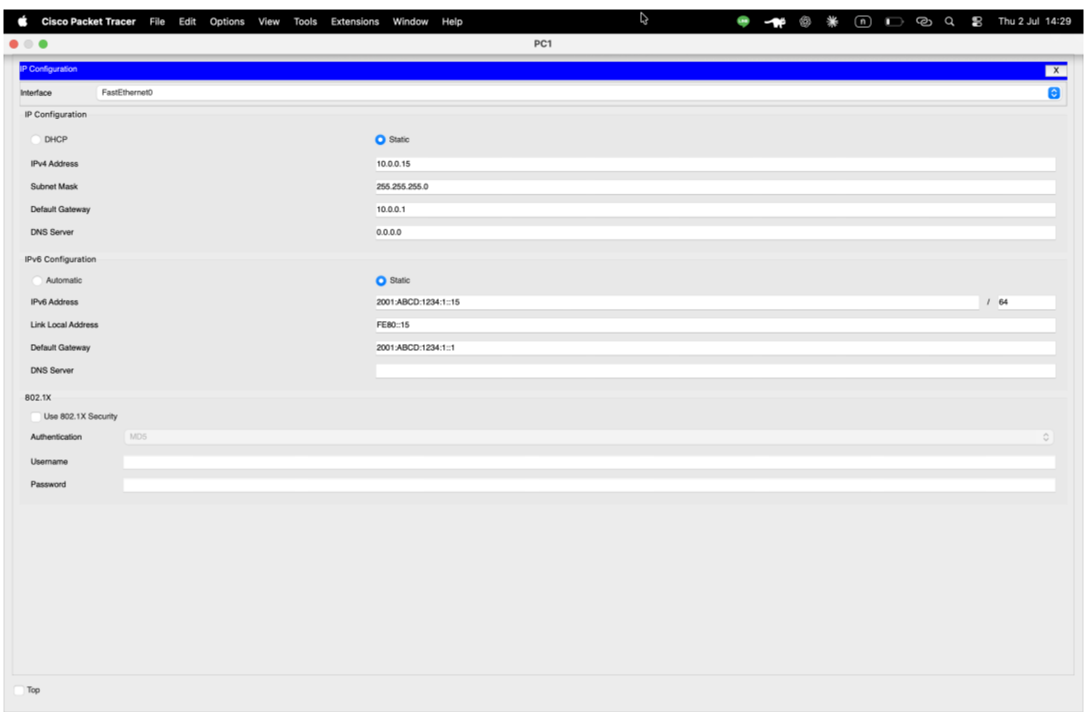
4. IPv6 Configuration Summary

4.1 Router NET_1_R1 — Interface Addressing

Interface	GUA	Link-Local	Status
G0/0/0	2001:ABCD:1234:1::1/64	FE80::1	up/up
G0/0/1	2001:ABCD:1234:2::1/64	FE80::1	up/up

4.2 PC IPv6 Addressing

Device	IPv6 Address (GUA)	Link-Local	Default Gateway
PC1	2001:ABCD:1234:1::15/64	FE80::15	2001:ABCD:1234:1::1
PC2	2001:ABCD:1234:1::16/64	FE80::16	2001:ABCD:1234:1::1
PC3	2001:ABCD:1234:1::17/64	FE80::17	2001:ABCD:1234:1::1
PC4	2001:ABCD:1234:1::18/64	FE80::18	2001:ABCD:1234:1::1
PC5	2001:ABCD:1234:1::19/64	FE80::19	2001:ABCD:1234:1::1
Admin1	2001:ABCD:1234:2::15/64	FE80::15	2001:ABCD:1234:2::1
Admin2	2001:ABCD:1234:2::16/64	FE80::16	2001:ABCD:1234:2::1



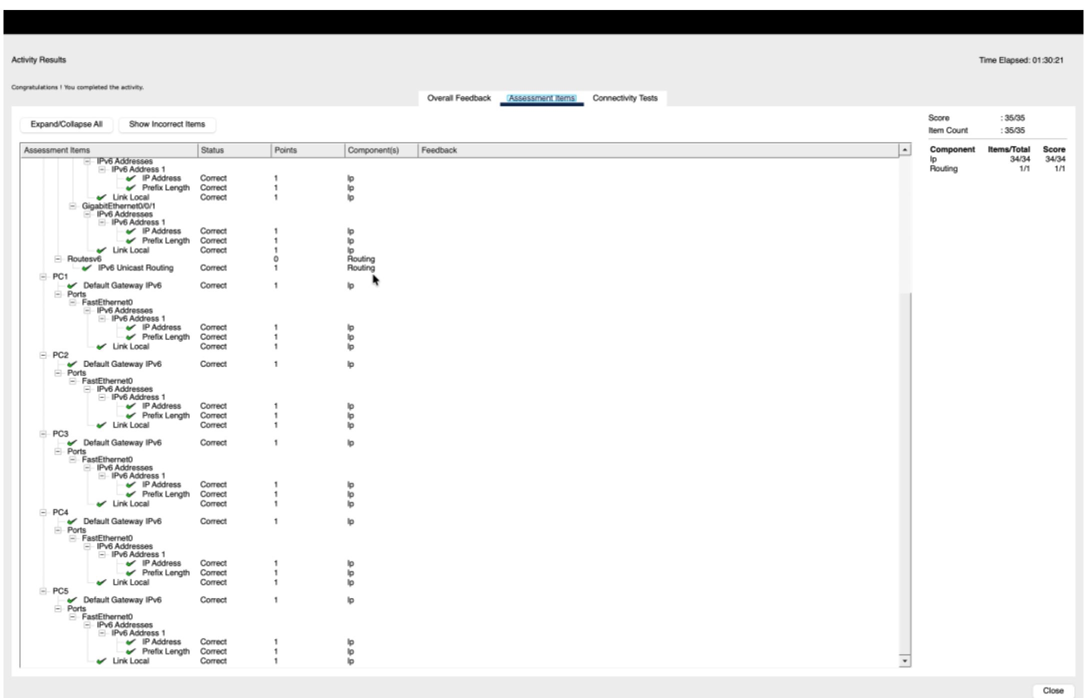
Screenshot: IP Configuration window — PC1 (IPv6 Address, Link Local, Default Gateway ครบถ้วน)

5. Connectivity Verification (PING Testing)

IPv6 connectivity was verified using PING between hosts on the same subnet and across subnets via NET_1_R1. Representative results are shown below; all address pairs matching the design table returned 0% packet loss.

Source	Destination	Destination IPv6	Result
PC1	PC2	2001:ABCD:1234:1::16	4/4 received – 0% loss
PC1	PC3	2001:ABCD:1234:1::17	4/4 received – 0% loss
PC1	PC4	2001:ABCD:1234:1::18	4/4 received – 0% loss
PC1	PC5	2001:ABCD:1234:1::19	4/4 received – 0% loss
PC1	Admin1	2001:ABCD:1234:2::15	4/4 received – 0% loss (cross-subnet via router)
PC1	Admin2	2001:ABCD:1234:2::16	4/4 received – 0% loss (cross-subnet via router)

Note: cross-subnet pings (PC-side to Admin-side) succeed because NET_1_R1 routes between the directly connected 2001:ABCD:1234:1::/64 and 2001:ABCD:1234:2::/64 networks — confirming that IPv6 routing is functioning correctly between G0/0/0 and G0/0/1.



Screenshot: Activity Results — Check Results (Score 35/35, ครบทุก component)

6. IPv6 Routing Table — NET_1_R1

Output of show ipv6 route on NET_1_R1, confirming both LAN prefixes are directly connected and the router owns the local host routes for each interface.

```
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

C   2001:ABCD:1234:1::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:ABCD:1234:1::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:ABCD:1234:2::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:ABCD:1234:2::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

7. Router Configuration — NET_1_R1 (show running-config)

```
Current configuration : 1116 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname NET_1_R1
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
ip cef
ipv6 unicast-routing
no ipv6 cef
!
no ip domain-lookup
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
 description Link to Room 100 LAN
 ip address 10.0.0.1 255.255.255.0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:ABCD:1234:1::1/64
!
interface GigabitEthernet0/0/1
 description Link to Admin and Wireless LAN
 ip address 11.0.0.1 255.255.255.0
 duplex auto
 speed auto
```

```
 ipv6 address FE80::1 link-local
 ipv6 address 2001:ABCD:1234:2::1/64
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.0.0 255.255.255.0 11.0.0.2
!
ip flow-export version 9
!
banner motd ^C Authorized Access Only ^C
!
line con 0
 password 7 0822455D0A16
 login
line aux 0
line vty 0 4
 password 7 0822455D0A16
 login
!
end
```

8. Additional Technical Information

8.1 Role of the Subnet Mask in IPv4 Addressing

A subnet mask divides an IPv4 address into a network portion and a host portion. Bits set to 1 in the mask represent the network portion; bits set to 0 represent the host portion. Performing a bitwise AND between an IP address and its subnet mask yields the network address, which routers use to decide whether to forward traffic locally or send it toward a gateway.

8.2 Determining Which Network a Host Belongs To

A host's network is found by performing a bitwise AND of its IP address with its subnet mask. For example, host 192.168.1.70 with mask 255.255.255.224 (/27) ANDs to network 192.168.1.64/27. Two hosts are on the same network only if this calculation produces the same network address for both.

8.3 Unicast, Multicast, and Broadcast Transmissions

- Unicast — one sender to one specific receiver (e.g., loading a webpage).
- Multicast — one sender to a defined group of interested receivers (e.g., video conferencing, streaming to a subscribed group).
- Broadcast — one sender to every host on the local network/subnet (e.g., ARP requests, DHCP Discover). Note: IPv6 has no broadcast; multicast is used instead.

8.4 Why IPv6 Addressing Is Needed

- IPv4's 32-bit address space (~4.3 billion addresses) is largely exhausted given the number of connected devices worldwide.
- IPv6 uses 128-bit addresses, providing an effectively unlimited supply for long-term growth.
- IPv6 was designed to reduce reliance on NAT, enabling simpler end-to-end connectivity.
- Built-in improvements include stateless address autoconfiguration (SLAAC), native security extensions, and the elimination of broadcast traffic.

8.5 Global Unicast Address (GUA) vs. Link-Local Address

Attribute	Global Unicast Address (GUA)	Link-Local Address
Scope	Routable across the internet	Valid only on the local link/segment
Prefix	2000::/3 (e.g., 2001:...)	Always FE80::/10
Routed by routers?	Yes	No — never forwarded off-link
Assignment	Static or via SLAAC/DHCPv6	Auto-generated whenever IPv6 is enabled

8.6 IPv6 Global Unicast Address Structure

A GUA is 128 bits, structured as follows:

```
| Global Routing Prefix | Subnet ID | Interface ID |
|        48 bits        |  16 bits  |   64 bits    |
```

- Global Routing Prefix (48 bits) — assigned by the ISP/registry; identifies the organisation on the internet.
- Subnet ID (16 bits) — used internally by the organisation to create subnets (this is the field used to separate the :1: and :2: segments in this design).
- Interface ID (64 bits) — identifies the specific host, either derived from the MAC address (EUI-64) or assigned manually, as done for PC1–PC5 and Admin1–Admin2 above.

8.7 How Host-to-Host Connectivity Is Tested — PING

PING uses ICMP (Internet Control Message Protocol) Echo Request/Reply messages. The source sends an Echo Request to the destination; if reachable, the destination returns an Echo Reply. The source measures round-trip time and reports the percentage of successful replies. No reply within the timeout period is reported as "Request timed out," indicating a connectivity problem.

8.8 PING vs. TRACERT

Attribute	PING	TRACERT / Traceroute
Purpose	Confirms whether a destination is reachable	Reveals the path (hops) packets take to reach a destination
Method	Sends ICMP Echo Request/Reply directly to the destination	Sends packets with increasing TTL values, forcing each intermediate router to respond
Output	Reachable/unreachable + response time	List of each hop's address and response time
Best used when...	You need a quick reachability check (e.g., after configuration)	PING fails and you need to isolate where along the path the failure occurs